Autonomous Decision Intelligence Engine for Enterprise Operations Using Multi-Agent AI

WORKFLOW IN ONE LINE (MEMORIZE THIS)

> Dataset → SQL ETL → Feature Engineering → ML Risk Prediction → Autonomous Decision Logic → Action Execution(Slack) → Power BI Oversight → Docker/K8s → AWS → CI/CD

PHASE 0 – Project Setup (Runs in Parallel Always)

What happens

- You create the repo
- You commit after every meaningful step
- You never work “locally only”

Tools

- Git
- GitHub

Artifacts

- Repository structure
- README.md (updated daily)

📌 **Git runs in parallel with everything else.**

PHASE 1 – Dataset → Database (Foundation)

Step 1: Dataset Upload

- Download **DataCo SMART Supply Chain dataset**
- Store it locally as CSV

Step 2: SQL ETL (Raw Load)

- Create MySQL schema
- Load CSV into `operations_events`
- Minimal cleaning (types, nulls)

Tools

- MySQL
- SQL
- Python (only for loading, optional)

Output

- Clean, queryable operational table

📌 This answers: **"What data do we have?"**

PHASE 2 – Feature Engineering (Data Science Begins)

What happens

- You derive features from raw data
- No ML yet

Examples

- Delivery delay
- SLA breach flag
- Priority score
- Time-based aggregates

Tools

- SQL (preferred)
- Python (pandas if needed)

Output

- Feature-ready dataset in MySQL

📌 This answers: **“What signals matter?”**

PHASE 3 – ML Models (Risk Agent)

What happens

- Pull feature data from MySQL
- Train ML models
- Generate predictions

Models

- Logistic Regression / Random Forest
- Target: late delivery / SLA breach

Tools

- Python
- scikit-learn
- SQLAlchemy (DB connection)

Output

- Risk scores
- Risk levels written to `risk_predictions`

📌 This answers: **“What will go wrong?”**

PHASE 4 – Autonomous Decision Logic (Decision Agent)

What happens

- Read risk predictions
- Apply business rules
- Decide actions automatically

Example

```
IFrisk_level= HIGH → EscalateIFrisk_level= MEDIUM → Monitor
```

Tools

- Python
- SQL

Output

- Decisions written to `decisions`

📌 This answers: **“What should we do?”**

PHASE 5 – Action Execution (Action Agent)

What happens

- Execute or simulate actions
- Log results

Examples

- Escalation logged
- Priority changed (simulated)
- Notification triggered (optional)

Tools

- Python
- SQL

Output

Slack Action Execution – Concrete Examples

Example 1: High-Risk Order Alert

```
🚨 High Risk Alert
Order ID: 56789
Predicted SLA Breach: YES
Risk Score: 0.86
Recommended Action: Escalate & Reprioritize
Triggered By: Autonomous Decision Engine
```

- Action records in `actions`

📌 This answers: **"What did the system do?"**

PHASE 6 – Power BI (Oversight Layer)

What happens

- Power BI connects to MySQL
- Dashboards read live decision data

Dashboards

- Operations overview
- Risk intelligence
- AI decisions
- End-to-end traceability

Tools

- Power BI Desktop
- Power BI Service (optional)

Output

- Interactive dashboards
- Auto-refresh on DB updates

📌 This answers: **"Can humans trust this AI?"**

PHASE 7 – Containerization & Deployment (DevOps)

What happens

- Package agents as Docker services
- Deploy locally or on AWS

Tools

- Docker
- Kubernetes (minimal)
- AWS EC2
- Terraform

Output

- Running system
- Infrastructure code

📌 This answers: **“Can this run in production?”**

PHASE 8 – CI/CD (Professional Touch)

What happens

- Jenkins builds images
- Runs tests
- Deploys automatically

Tools

- Jenkins
- GitHub Webhooks

Output

- Automated pipeline
- Zero manual deployment

📌 This answers: **“Is this enterprise-ready?”**